



Lecture 2.2: Overhead Transmission Lines

EEE 321: Power System I

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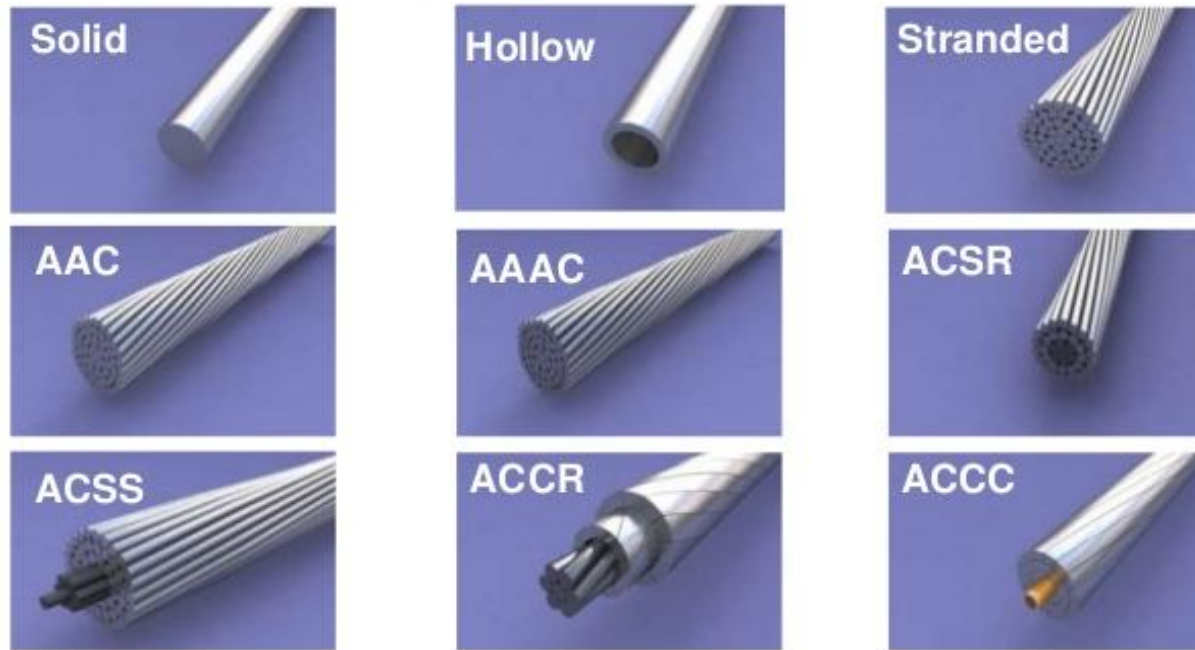
Presentation Outline

- ❑ Overhead Transmission Lines- Aluminium Conductors
- ❑ Bundled Conductors in Overhead Transmission
- ❑ Circuit Parameters of Overhead Transmission Lines
- ❑ Inductance
- ❑ Capacitance
- ❑ Problems

Overhead Transmission Line- Aluminium Conductors

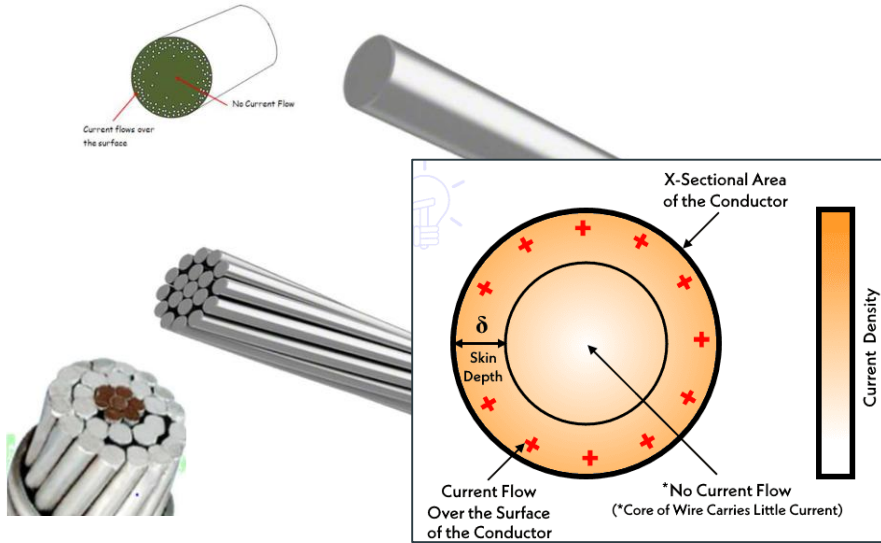
Aluminium is cheap and light as compared to copper but it has much smaller conductivity and tensile strength. Due to low tensile strength, aluminium conductors produce greater sag. This prohibits their use for larger spans and makes them unsuitable for long distance transmission. In order to increase the tensile strength, the aluminium conductor is reinforced with a core of galvanised steel wires. The composite conductor thus obtained is known as steel cored aluminium and is abbreviated as A.C.S.R. (aluminium conductor steel reinforced).

Types of Aluminium Conductors



- All Aluminium Conductors (AAC)
- All Aluminium Alloy Conductors (AAAC)
- Aluminium Conductor Steel Reinforced (ACSR)/Composite Stranded Conductor
- Aluminium Conductor Alloy Reinforced (ACAR)
- Aluminum Conductor Composite Reinforced (ACCR)
- Aluminium Conductor Composite Core (ACCC)

Overhead Transmission Line- Aluminium Conductors



To reduce the Skin Effect:

- ❑ Current in a solid conductor in AC tends to concentrate on the periphery due to **skin effect**. So the conductor in reality behaves like a hollow pipe and inner bulk of the cross section carries no current.
- ❑ Stranding divides the conductor into a number of conductors in parallel, thereby effective current carrying area and current capacity is increased. This is one of the major reasons for stranded wires being used everywhere.

Why do transmission and distribution lines use stranded conductors?

To enhance flexibility:

Solid conductors, especially larger ones, can be quite rigid, making them difficult to install and prone to damage from bending and vibration. Stranded conductors, with their multiple strands, offer much greater flexibility, allowing for easier installation and handling, particularly on long spans.

To increase durability:

Overhead lines are subjected to wind and temperature changes that can cause significant stress and vibration. The individual strands in a stranded conductor are less susceptible to fatigue and breakage than a single solid conductor.

To improve Mechanical Strength:

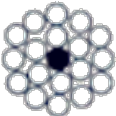
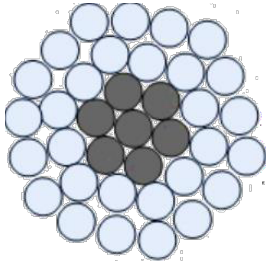
While the primary reason for stranding is flexibility, the steel core in some stranded conductors like ACSR (Aluminum Conductor Steel Reinforced) provides significant mechanical strength to withstand tension and other stresses.

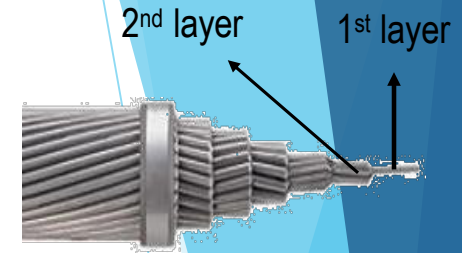
To reduce overall expenses:

While the initial cost of stranding can be slightly higher, the reduced maintenance and longer lifespan often make it a more cost-effective solution in the long run.

Overhead Transmission Line- Aluminium Conductors

Conductor Layer, Strand and Diameter

Total no. of layers, x	Total no. of strands, $N = 3x^2 - 3x + 1$	Diameter of Conductor, $D = (2x - 1)d$ d=strand diameter	Cross sectional view of the conductor
2	7 (1+6)	3d	
3	19 (1+6+12)	5d	
4	37 (1+6+12+18)	7d	



Alternate layer of strands spiraled in opposite directions



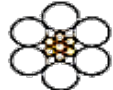
Alternate layer of strands are laid in spiraled form in opposite directions to hold the strands together and make the outer radius of one layer coincide with the inner radius of the next.

Overhead Transmission Line- Aluminium Conductors

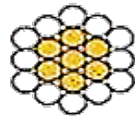
ACSR Types/Ratio of Aluminum and Steel in ACSR [Al+St]



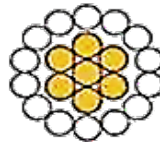
6+1



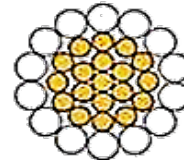
6+7(1+6)



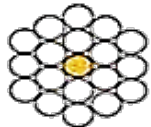
12+7(1+6)



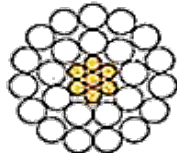
14+7(1+6)



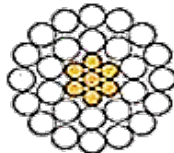
14+19(1+6+12)



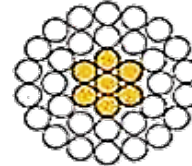
18(6+12)+1



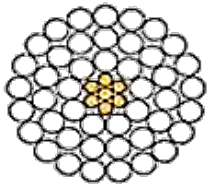
24(9+15)+7(1+6)



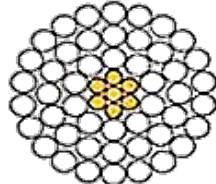
26(10+16)+7(1+6)



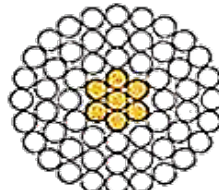
30(12+18)+7(1+6)



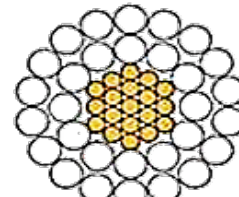
45(9+15+21)+7(1+6)



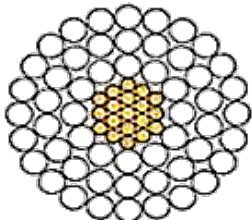
48(10+16+22)+7(1+6)



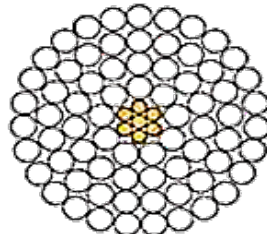
54(12+18+24)+7(1+6)



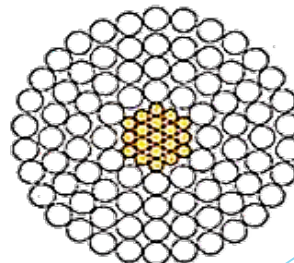
30(12+18)+19(1+6+12)



54(12+18+24)+19(1+6+12)



72(9+15+21+27)+7(1+6)



84(12+18+24+30)+19(1+6+12)

Overhead Transmission Line- Aluminium Conductors

ACSR Conductors Animal Names Data Sheet

Animal Name	Code	Aluminum Strands	Steel Strands	Total Strands	Diameter (Inches)
Ant	ANT	6	1	7	0.292
Squirrel	SQ	6	1	7	0.346
Dog	DOG	12	7	19	0.563
Weasel	WE	18	1	19	0.376
Rabbit	RB	26	7	33	0.691
Ferret	FR	30	7	37	0.726
Beaver	BV	42	7	49	0.903
Raccoon	RC	54	7	61	1.024
Fox	FX	61	19	80	1.224
Moose	MS	72	19	91	1.354
Deer	DR	84	19	103	1.474
Elk	EK	104	37	141	1.766
Bison	BS	126	37	163	1.976
Bear	BR	147	37	184	2.156
Zebra	ZB	168	37	205	2.328

In the electrical industry, ACSR (Aluminum Conductor Steel Reinforced) cables use codenames based on birds and animals. Bird names are predominantly used in North America (per the ASTM standards), while other land and wild animal names are widely adopted in the UK, Europe, and parts of Asia to quickly identify specific cross-sections, strand counts, and tensile strengths

Ref: <https://prabhapower.com/acsr-conductors-animal-inspired-name-sizes/>

Overhead Transmission Line- Aluminium Conductors

ACSR Conductors Data Sheet (Following Bird's names)

Code word	Aluminum area, cmil	Stranding Al/St	Layers of aluminum	Outside diameter, in	Resistance			GMR D_s , ft	Reactance per conductor 1-ft spacing, 60 Hz	
					Dc, 20°C, $\Omega/1,000$ ft	Ac, 60 Hz			Inductive X_a , Ω/mi	Capacitive X'_a , $\text{M}\Omega\cdot\text{mi}$
						20°C, Ω/mi	50°C, Ω/mi			
Waxwing	266,800	18/1	2	0.609	0.0646	0.3488	0.3831	0.0198	0.476	0.1090
Partridge	266,800	26/7	2	0.642	0.0640	0.3452	0.3792	0.0217	0.465	0.1074
Ostrich	300,000	26/7	2	0.680	0.0569	0.3070	0.3372	0.0229	0.458	0.1057
Merlin	336,400	18/1	2	0.684	0.0512	0.2767	0.3037	0.0222	0.462	0.1055
Linnet	336,400	26/7	2	0.721	0.0507	0.2737	0.3006	0.0243	0.451	0.1040
Oriole	336,400	30/7	2	0.741	0.0504	0.2719	0.2987	0.0255	0.445	0.1032
Chickadee	397,500	18/1	2	0.743	0.0433	0.2342	0.2572	0.0241	0.452	0.1031
Ibis	397,500	26/7	2	0.783	0.0430	0.2323	0.2551	0.0264	0.441	0.1015
Pelican	477,000	18/1	2	0.814	0.0361	0.1957	0.2148	0.0264	0.441	0.1004
Flicker	477,000	24/7	2	0.846	0.0359	0.1943	0.2134	0.0284	0.432	0.0992
Hawk	477,000	26/7	2	0.858	0.0357	0.1931	0.2120	0.0289	0.430	0.0988
Hen	477,000	30/7	2	0.883	0.0355	0.1919	0.2107	0.0304	0.424	0.0980
Osprey	556,500	18/1	2	0.879	0.0309	0.1679	0.1843	0.0284	0.432	0.0981
Parakeet	556,500	24/7	2	0.914	0.0308	0.1669	0.1832	0.0306	0.423	0.0969
Dove	556,500	26/7	2	0.927	0.0307	0.1663	0.1826	0.0314	0.420	0.0965
Rook	636,000	24/7	2	0.977	0.0269	0.1461	0.1603	0.0327	0.415	0.0950
Grosbeak	636,000	26/7	2	0.990	0.0268	0.1454	0.1596	0.0335	0.412	0.0946
Drake	795,000	26/7	2	1.108	0.0215	0.1172	0.1284	0.0373	0.399	0.0912
Tern	795,000	45/7	3	1.063	0.0217	0.1188	0.1302	0.0352	0.406	0.0925
Rail	954,000	45/7	3	1.165	0.0181	0.0997	0.1092	0.0386	0.395	0.0897
Cardinal	954,000	54/7	3	1.196	0.0180	0.0988	0.1082	0.0402	0.390	0.0890
Ortolan	1,033,500	45/7	3	1.213	0.0167	0.0924	0.1011	0.0402	0.390	0.0885
Bluejay	1,113,000	45/7	3	1.259	0.0155	0.0861	0.0941	0.0415	0.386	0.0874
Finch	1,113,000	54/19	3	1.293	0.0155	0.0856	0.0937	0.0436	0.380	0.0866
Bittern	1,272,000	45/7	3	1.345	0.0136	0.0762	0.0832	0.0444	0.378	0.0855
Pheasant	1,272,000	54/19	3	1.382	0.0135	0.0751	0.0821	0.0466	0.372	0.0847
Bobolink	1,431,000	45/7	3	1.427	0.0121	0.0684	0.0746	0.0470	0.371	0.0837
Plover	1,431,000	54/19	3	1.465	0.0120	0.0673	0.0735	0.0494	0.365	0.0829
Lapwing	1,590,000	45/7	3	1.502	0.0109	0.0623	0.0678	0.0498	0.364	0.0822
Falcon	1,590,000	54/19	3	1.545	0.0108	0.0612	0.0667	0.0523	0.358	0.0814
Bluebird	2,156,000	84/19	4	1.762	0.0080	0.0476	0.0515	0.0586	0.344	0.0776

[†] Most used multilayer sizes.

[‡] Data, by permission, from Aluminum Association, *Aluminum Electrical Conductor Handbook*, 2nd ed., Washington, D.C., 1982.

Overhead Transmission Line- Aluminium Conductors

To Increase conductivity and tensile strength and reduce corona effect

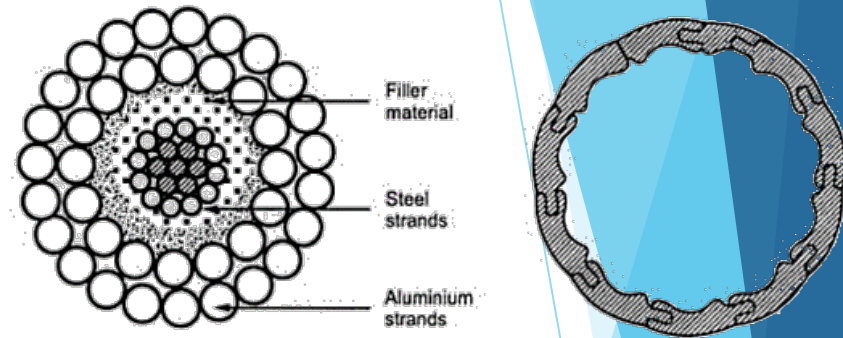
❑ Expanded ACSR

Expanded ACSR conductors are made by adding plastic or paper between the inner steel strands from the outer aluminum strands. The paper gives a larger diameter. The corona effect, which is a discharge of electricity in the air around high-voltage conductors, is inversely proportional to the conductor's diameter. A larger diameter means a lower electric field intensity at the conductor's surface, making it less likely to ionize the surrounding air and cause corona discharge.

Expanded ACSR is used for some extra-high-voltage (EHV) lines.

Some voltage classifications are as follows:

- HV: 115 kV, 138 kV, 230 kV
- EHV: 345 kV, 500 kV, 765 kV
- UHV: 1000 kV, 1500 kV



Overhead Transmission Line- Aluminium Conductors

Aluminium Conductor Composite Core (ACCC)

ACCC conductor (Aluminum Conductor Composite Core) is a high capacity, low sag conductor which consists of a carbon fiber composite core encased in a protective fiberglass sheath that is helically wrapped with conductive aluminum strands.

Greater Strength & Reduced Sag

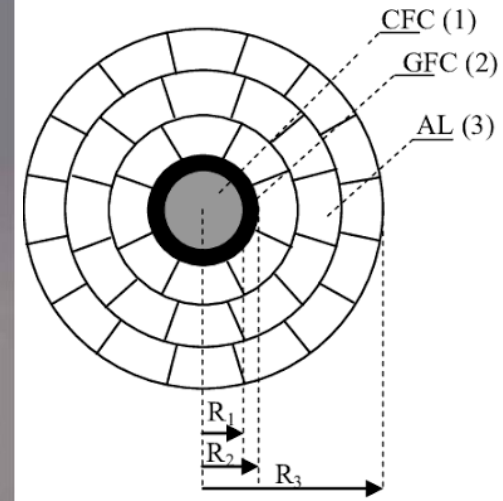
Increased Spans on Fewer / Shorter Structures

Twice the Capacity of AAC, ACSR and Others

Reduced Line Losses by 25% to 40%

Lower Overall Lifecycle Costs

Outstanding Product Warranty



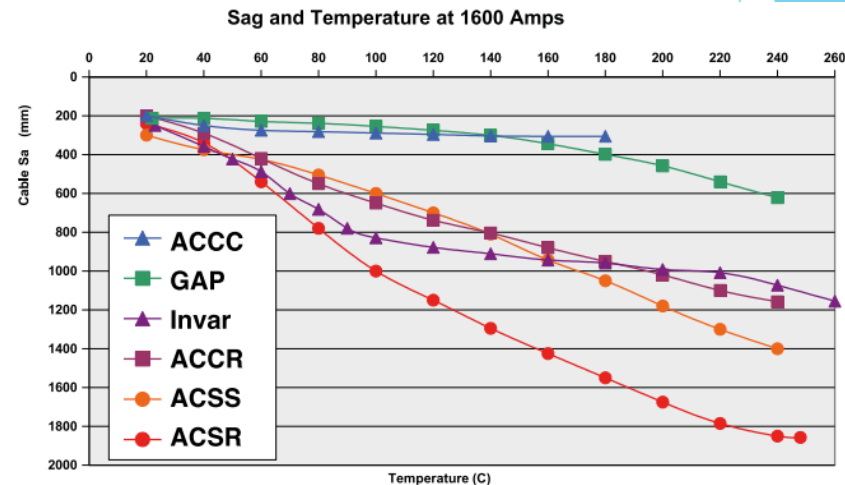
The core is made of carbon fiber epoxy composite (CFC). The next layer protects aluminum against corrosion from carbon fibers. The layer material is a glass fiber epoxy composite (GFC). The additional task of GFC is to increase the strength and flexibility of the core. The last area is the current conductor. The conductor is twisted from soft and annealed aluminum trapezoidal strands.

Overhead Transmission Line- Aluminium Conductors

Aluminium Conductor Composite Core (ACCC)

ACCC technology is based on replacing the heavy steel core of traditional electrical transmission conductors with a high strength, light weight carbon fiber composite core and is applicable for all voltages

The lower Coefficient of Thermal Expansion creates less sag at high temperatures and is resists degradation from vibration, corrosion, ultraviolet radiation, corona, chemical and thermal oxidation, but most importantly cyclic load fatigue.



Economics of Efficiency

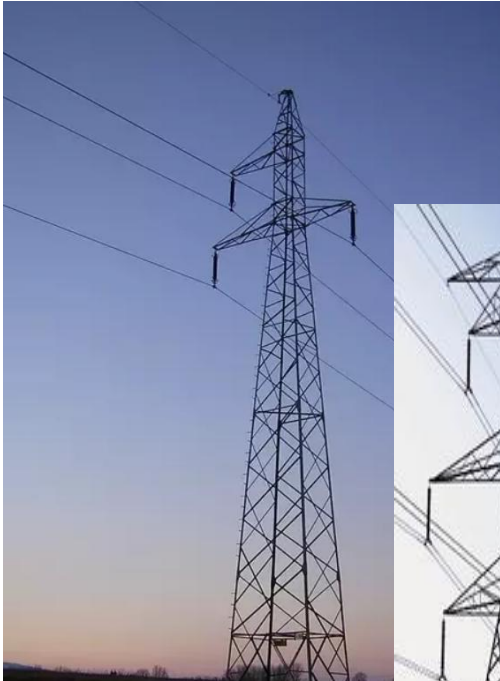
Investments in the ACCC® conductor will significantly reduce line losses and provide a higher Internal Rate of Return (IRR).

	Peak Amps	Temperature at Peak Amps (°C)	Load Factor	MVA	Annual Line Losses (MWh)	Line Loss Reduction	Value of Reduction (at \$50/MWh)	Value of Reduction per Lineal Conductor (meter)
ACSR	1,000	91	53%	398	75,980	----	----	----
ACCC®	1,000	77	53%	398	55,630	20,350	\$1,017,500	\$3.39
ACSS	1,600	194	53%	637	251,998	----	----	----
ACCC®	1,600	156	53%	637	179,022	72,976	\$3,648,800	\$12.16

Bundled Conductors in Overhead Transmission

More than one conductor per phase (Two, three, four or more). For transmission of power beyond 200kV bundled conductors are preferred.

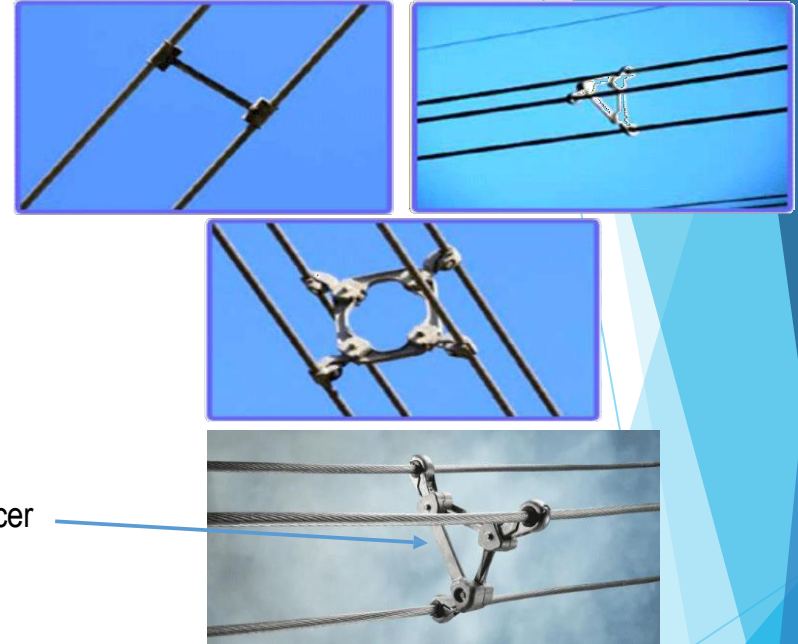
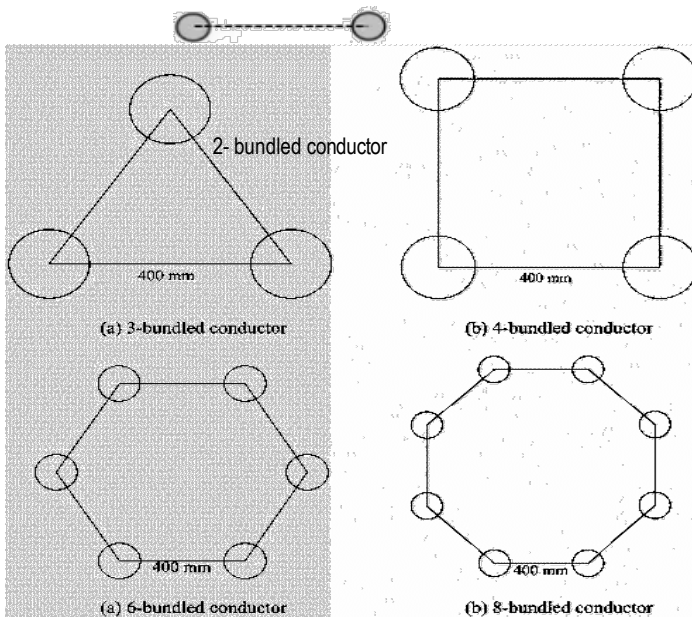
Can you identify the number of conductors per phase?



Bundled Conductors in Overhead Transmission

Bundle arrangement

The conductors of a bundle are separated at constant intervals **with spacer dampers** to connect the conductors in parallel. They also prevent clashing of the conductors and prevent them from swing in the wind.



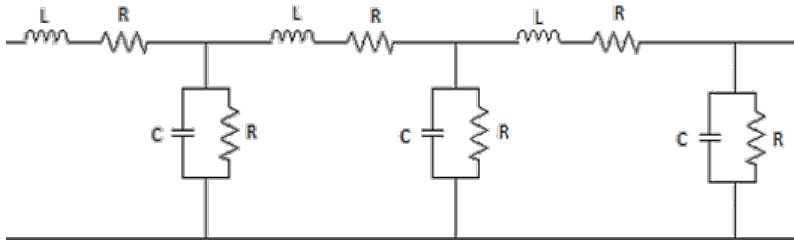
Spacer

Advantages

- **Reduce line inductance** which in turn increases power transfer capability of line.
- **Reduce corona discharge**. Using several conductors nearby instead of one conductor, forming a bundled conductor **with larger diameter** which leads to a reduction of voltage gradient and hence the possibility of corona formation. Reduction in the formation of corona discharge leads to **less power loss and hence improved transmission efficiency of the line**.
- **Reduction in communication line** interference due to reduction in corona.

Spacer dampers are crucial components in bundled conductor transmission lines, serving to maintain conductor spacing and dampen vibrations caused by wind and other dynamic forces.

Circuit Parameters of Overhead Transmission Lines



An equivalent circuit of line can be represented by four electrical parameters:

Series resistance (R)

Depends on the physical composition of the conductor at a given temperature and causes actual power loss in the conductor. **It is important for transmission efficiency evaluation and economic studies.**

Series inductance (L)

Due to the magnetic properties of the conductor. Power transmission capacity of T. L is basically governed by inductance.

Shunt conductance (G)

Due to the leakage current flowing across insulators. **The leakage currents are negligible compared to the transmission line current.**

Shunt capacitance (C)

Due to the electrical field properties between conductors

Electrical performances of Transmission Line i.e. efficiency and voltage regulation depend on the value of line parameters. Therefore, for designing a line, knowledge of calculating and setting the values of these parameters is very important.

Circuit Parameters of Overhead Transmission Lines

RESISTANCE

The dc resistance of a conductor at a specified temperature T is

$$R_{dc,T} = \frac{\rho_T l}{A} \quad \Omega \quad (4.2.1)$$

where ρ_T = conductor resistivity at temperature T

l = conductor length

A = conductor cross-sectional area

Conductor resistance depends on the following factors:

1. Spiraling 2. Temperature 3. Frequency (“skin effect”) 4. Current magnitude—magnetic conductors

For stranded conductors, alternate layers of strands are spiraled in opposite directions to hold the strands together. Spiraling makes the strands 1 or 2% longer than the actual conductor length. As a result, the dc resistance of a stranded conductor is 1 or 2% larger than that calculated from (4.2.1) for a specified conductor length.

Resistivity of conductor metals varies linearly over normal operating temperatures according to

$$\rho_{T2} = \rho_{T1} \left(\frac{T_2 + T}{T_1 + T} \right) \quad (4.2.3)$$

where ρ_{T2} and ρ_{T1} are resistivities at temperatures T_2 and T_1 °C, respectively. T is a temperature constant that depends on the conductor material, and is listed in Table 4.3.

The ac resistance or *effective* resistance of a conductor is

$$R_{ac} = \frac{P_{loss}}{|I|^2} \quad \Omega \quad (4.2.4)$$

For dc, the current distribution is uniform throughout the conductor cross section, and (4.2.1) is valid. However, for ac, the current distribution is nonuniform.

Circuit Parameters of Overhead Transmission Lines

RESISTANCE

TABLE 4.3: % Conductivity, resistivity, and temperature constant of conductor metal

Material	% Conductivity	$\rho_{20^{\circ}\text{C}}$	T
		Resistivity at 20 °C	Temperature Constant
		$\Omega\text{m} \times 10^{-8}$	$^{\circ}\text{C}$
Copper:			
Annealed	100%	1.72	234.5
Hard-drawn	97.3%	1.77	241.5
Aluminum			
Hard-drawn	61%	2.83	228.1
Brass	20–27%	6.4–8.4	480
Iron	17.2%	10	180
Silver	108%	1.59	243
Sodium	40%	4.3	207
Steel	2–14%	12–88	180–980

As frequency increases, the current in a solid cylindrical conductor tends to crowd toward the conductor surface, with smaller current density at the conductor center. This phenomenon is called skin effect. A conductor with a large radius can even have an oscillatory current density versus the radial distance from the conductor centre.

With increasing frequency, conductor loss increases, which, from (4.2.4), causes the ac resistance to increase. At power frequencies (60 Hz), the ac resistance is at most a few percent higher than the dc resistance. Conductor manufacturers normally provide dc, 50-Hz, and 60-Hz conductor resistance based on test data

Circuit Parameters of Overhead Transmission Lines

CONDUCTANCE

- ❑ Conductance accounts for real power loss between conductors or between conductors and ground. For overhead lines, this power loss is due to leakage currents at insulators and to corona.
- ❑ Insulator leakage current depends on the amount of dirt, salt, and other contaminants that have accumulated on insulators, as well as on meteorological factors, particularly the presence of moisture.
- ❑ Corona occurs when a high value of electric field strength at a conductor surface causes the air to become electrically ionized and to conduct. The real power loss due to corona, called corona loss, depends on meteorological conditions, particularly rain, and on conductor surface irregularities.
- ❑ Losses due to insulator leakage and corona are usually small compared to conductor I^2R loss.
- ❑ Conductance is usually neglected in power system studies because it is a very small component of the shunt admittance.

Circuit Parameters of Overhead Transmission Lines

INDUCTANCE: SINGLE-PHASE TWO-WIRE LINE

SOLID CYLINDRICAL CONDUCTOR

Figure 4.9(a) shows a single-phase two-wire line consisting of two solid cylindrical conductors x and y . Conductor x with radius r_x carries phasor current $I_x = I$ referenced out of the page. Conductor y with radius r_y carries return current $I_y = -I$. Since the sum of the two currents is zero, (4.4.30) is valid, from which the total flux linking conductor x is

$$\begin{aligned}\lambda_x &= 2 \times 10^{-7} \left(I_x \ln \frac{1}{D_{xx}} + I_y \ln \frac{1}{D_{xy}} \right) \\ &= 2 \times 10^{-7} \left(I \ln \frac{1}{r'_x} - I \ln \frac{1}{D} \right) \\ &= 2 \times 10^{-7} I \ln \frac{D}{r'_x} \quad \text{Wb-t/m}\end{aligned}$$

where $r'_x = e^{-1/4} r_x = 0.7788 r_x$.

The inductance of conductor x is then

$$L_x = \frac{\lambda_x}{I_x} = \frac{\lambda_x}{I} = 2 \times 10^{-7} \ln \frac{D}{r'_x} \quad \text{H/m per conductor}$$

Similarly, the total flux linking conductor y is

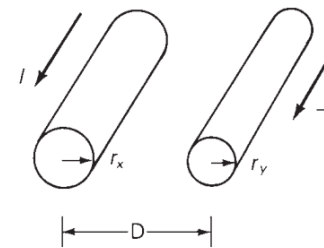
$$\begin{aligned}\lambda_y &= 2 \times 10^{-7} \left(I_x \ln \frac{1}{D_{yx}} + I_y \ln \frac{1}{D_{yy}} \right) \\ &= 2 \times 10^{-7} \left(I \ln \frac{1}{D} - I \ln \frac{1}{r'_y} \right) \\ &= -2 \times 10^{-7} I \ln \frac{D}{r'_y}\end{aligned}$$



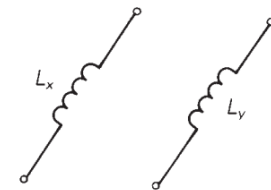
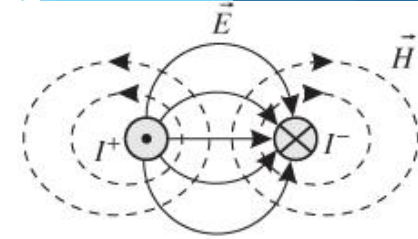
$$L_y = \frac{\lambda_y}{I_y} = \frac{\lambda_y}{-I} = 2 \times 10^{-7} \ln \frac{D}{r'_y} \quad \text{H/m per conductor} \quad (4.5.4)$$

FIGURE 4.9

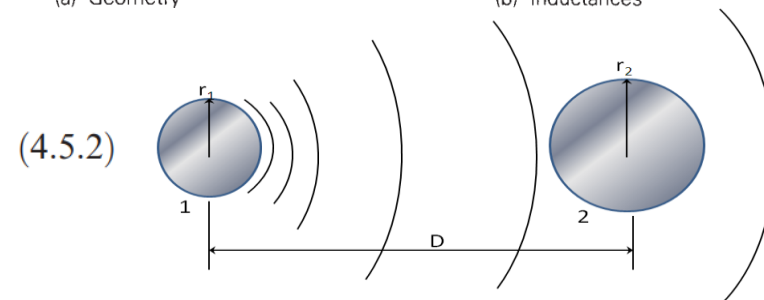
Single-phase two-wire line



(a) Geometry



(b) Inductances



Circuit Parameters of Overhead Transmission Lines

INDUCTANCE: SINGLE-PHASE TWO-WIRE LINE

SOLID CYLINDRICAL CONDUCTOR

The total inductance of the single-phase circuit, also called *loop inductance*, is

$$\begin{aligned} L &= L_x + L_y = 2 \times 10^{-7} \left(\ln \frac{D}{r'_x} + \ln \frac{D}{r'_y} \right) \\ &= 2 \times 10^{-7} \ln \frac{D^2}{r'_x r'_y} \\ &= 4 \times 10^{-7} \ln \frac{D}{\sqrt{r'_x r'_y}} \quad \text{H/m per circuit} \end{aligned} \quad (4.5.5)$$

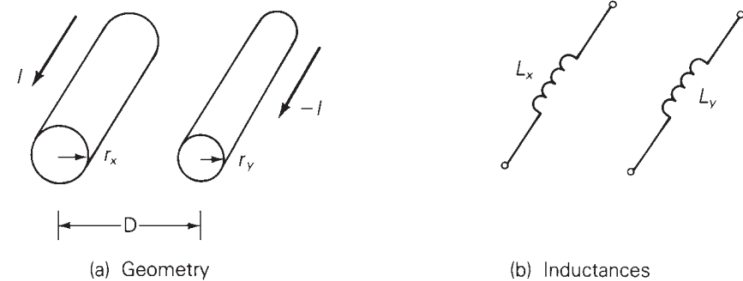
Also, if $r'_x = r'_y = r'$, the total circuit inductance is

$$L = 4 \times 10^{-7} \ln \frac{D}{r'} \quad \text{H/m per circuit} \quad (4.5.6)$$

The inductances of the single-phase two-wire line are shown in Figure 4.9(b).

Loop inductance is double the inductance per conductor in a single phase line.

FIGURE 4.9
Single-phase two-wire line



Circuit Parameters of Overhead Transmission Lines

INDUCTANCE: THREE-PHASE THREE-WIRE LINE WITH EQUAL PHASE SPACING

Figure 4.10(a) shows a three-phase three-wire line consisting of three solid cylindrical conductors a , b , c , each with radius r , and with equal phase spacing D between any two conductors. To determine inductance, assume balanced positive-sequence currents I_a , I_b , I_c that satisfy $I_a + I_b + I_c = 0$. Then (4.4.30) is valid and the total flux linking the phase a conductor is

$$\begin{aligned}\lambda_a &= 2 \times 10^{-7} \left(I_a \ln \frac{1}{r'} + I_b \ln \frac{1}{D} + I_c \ln \frac{1}{D} \right) \\ &= 2 \times 10^{-7} \left[I_a \ln \frac{1}{r'} + (I_b + I_c) \ln \frac{1}{D} \right]\end{aligned}$$

Using $(I_b + I_c) = -I_a$,

$$\begin{aligned}\lambda_a &= 2 \times 10^{-7} \left(I_a \ln \frac{1}{r'} - I_a \ln \frac{1}{D} \right) \\ &= 2 \times 10^{-7} I_a \ln \frac{D}{r'} \quad \text{Wb-t/m}\end{aligned}$$

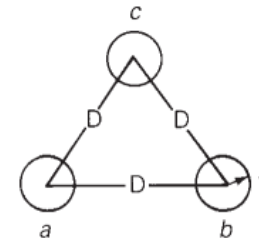
The inductance of phase a is then

$$L_a = \frac{\lambda_a}{I_a} = 2 \times 10^{-7} \ln \frac{D}{r'} \quad \text{H/m per phase}$$

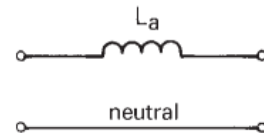
Due to symmetry, the same result is obtained for $L_b = \lambda_b/I_b$ and for $L_c = \lambda_c/I_c$. However, only one phase need be considered for balanced three-phase operation of this line, since the flux linkages of each phase have equal magnitudes and 120° displacement. The phase inductance is shown in Figure 4.10(b).

FIGURE 4.10

Three-phase three-wire line with equal phase spacing



(a) Geometry



(b) Phase inductance

(4.5.8)

SOLID CYLINDRICAL CONDUCTOR

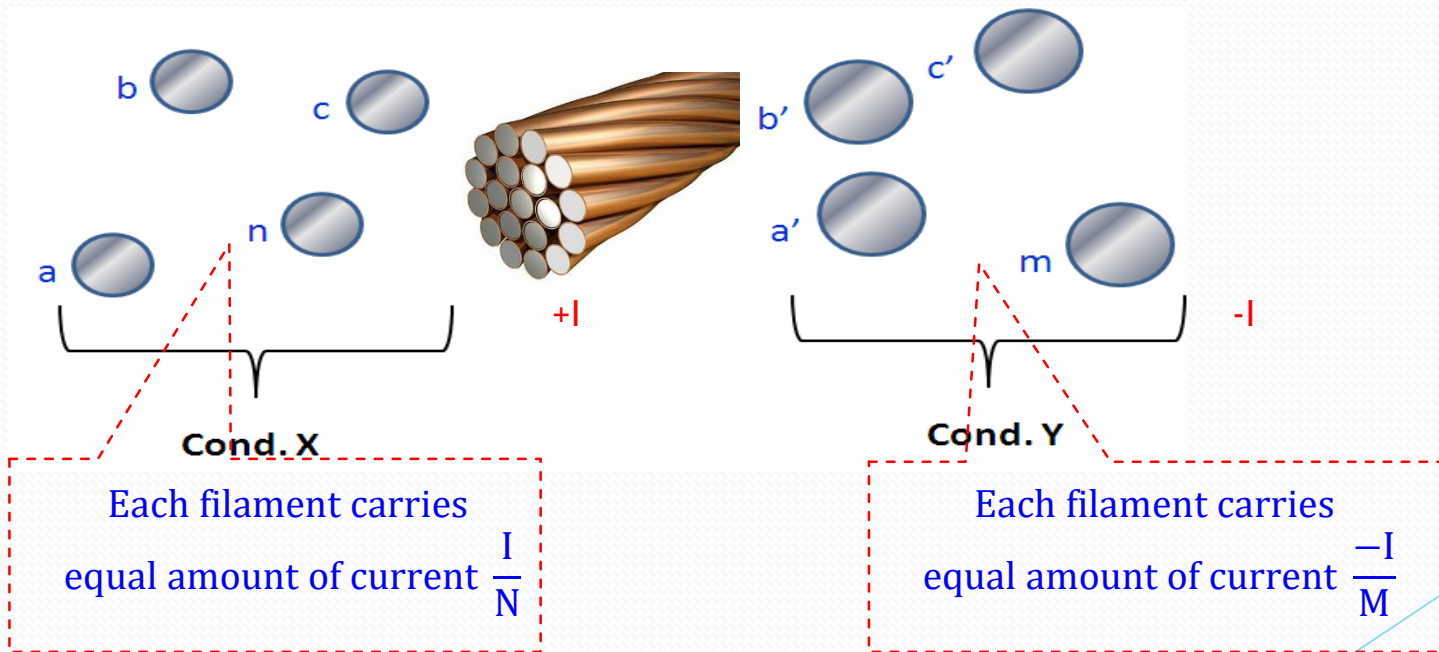
(4.5.9)

Circuit Parameters of Overhead Transmission Lines

INDUCTANCE: SINGLE-PHASE TWO-WIRE LINE

COMPOSITE CONDUCTORS

Composite conductor composed of two or more elements or strands **electrically in parallel**. The figure shows a **single phase having two conductors**. Each conductor, forming each side of the line, consists of indefinite number of strands with arbitrary arrangement. **The only restrictions are that the parallel filaments are cylindrical and share the current equally.**



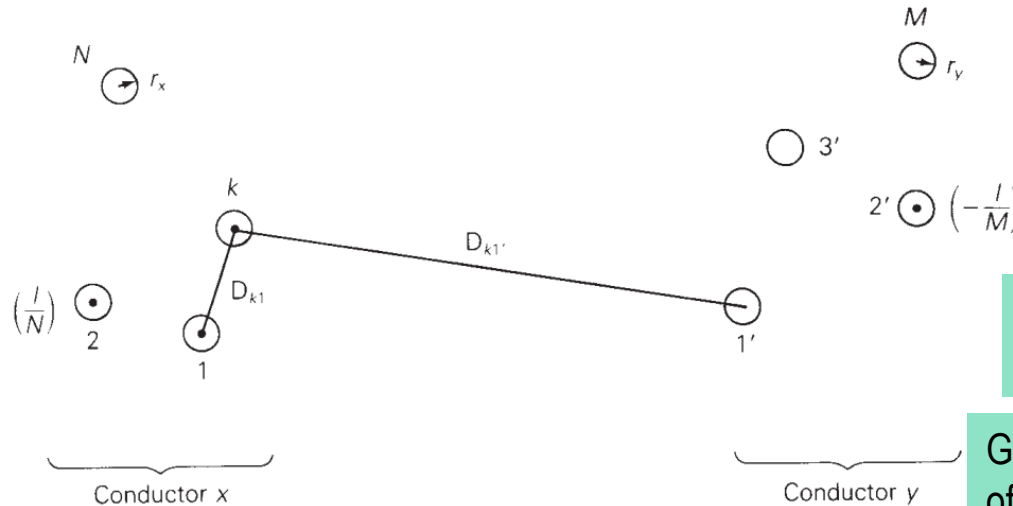
Circuit Parameters of Overhead Transmission Lines

INDUCTANCE: SINGLE-PHASE TWO-WIRE LINE

COMPOSITE CONDUCTORS

FIGURE 4.11

Single-phase two-conductor line with composite conductors



the inductance of conductor x, $L_x = \frac{\lambda_x}{I}$, can be written as

$$L_x = 2 \times 10^{-7} \ln \frac{D_{xy}}{D_{xx}} \text{ H/m per conductor} \quad (4.6.5)$$

Similarly, for conductor y,

$$L_y = 2 \times 10^{-7} \ln \frac{D_{xy}}{D_{yy}} \text{ H/m per conductor} \quad (4.6.8)$$

The total inductance L of the single-phase circuit is

$$L = L_x + L_y \text{ H/m per circuit} \quad (4.6.10)$$

Geometric Mean Distance (GMD) between conductors

Geometric Mean Radius (GMR) of conductor

where

$$D_{xy} = \sqrt[NM]{\prod_{k=1}^N \prod_{m=1'}^M D_{km}}$$

$$D_{xx} = \sqrt{N^2 \prod_{k=1}^N \prod_{m=1}^N D_{km}}$$

$$D_{yy} = \sqrt{M^2 \prod_{k=1'}^M \prod_{m=1'}^M D_{km}}$$

Conductor y

Conductor x

Circuit Parameters of Overhead Transmission Lines

INDUCTANCE: THREE-PHASE THREE-WIRE LINE WITH EQUAL/UNEQUAL PHASE SPACING

To calculate inductance for three-phase lines with stranded conductors and equal phase spacing, r' is replaced by the conductor GMR in (4.5.9). If the spacings between phases are unequal, then balanced positive-sequence flux linkages are not obtained from balanced positive-sequence currents. Instead, unbalanced flux linkages occur, and the phase inductances are unequal.

However, balance can be restored by exchanging the conductor positions along the line a technique called transposition

COMPOSITE CONDUCTORS

FIGURE 4.13

Completely transposed three-phase line

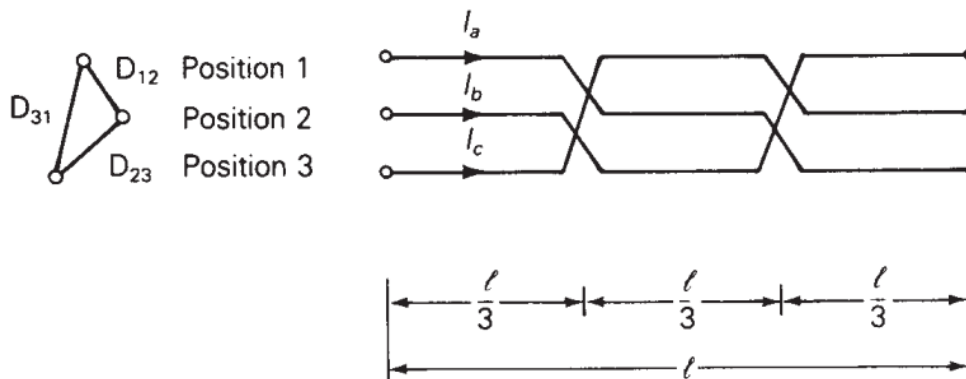


Figure 4.13 shows a completely transposed three-phase line. The line is transposed at two locations such that each phase occupies each position for one-third of the line length. Conductor positions are denoted 1, 2, 3 with distances D_{12} , D_{23} , D_{31} between positions.

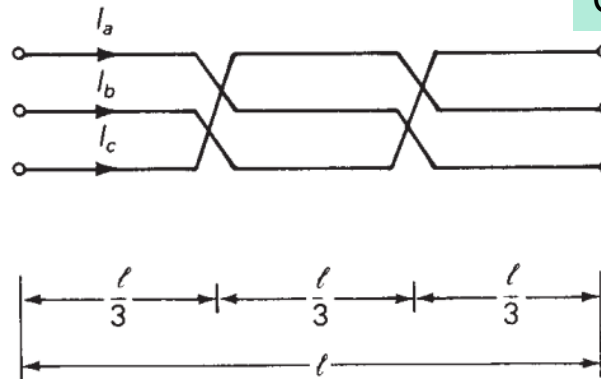
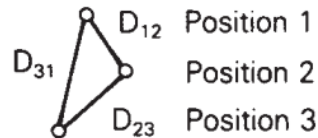
The conductors are identical, each with GMR denoted D_s . To calculate inductance of this line, assume balanced positive-sequence currents I_a, I_b, I_c , for which $I_a + I_b + I_c = 0$.

Circuit Parameters of Overhead Transmission Lines

INDUCTANCE: THREE-PHASE THREE-WIRE LINE WITH EQUAL/UNEQUAL PHASE SPACING

FIGURE 4.13

Completely transposed three-phase line



COMPOSITE CONDUCTORS

Only one phase need be considered for balanced three-phase operation of a completely transposed three-phase line. Defining

$$D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}} \quad (4.6.17)$$

we have

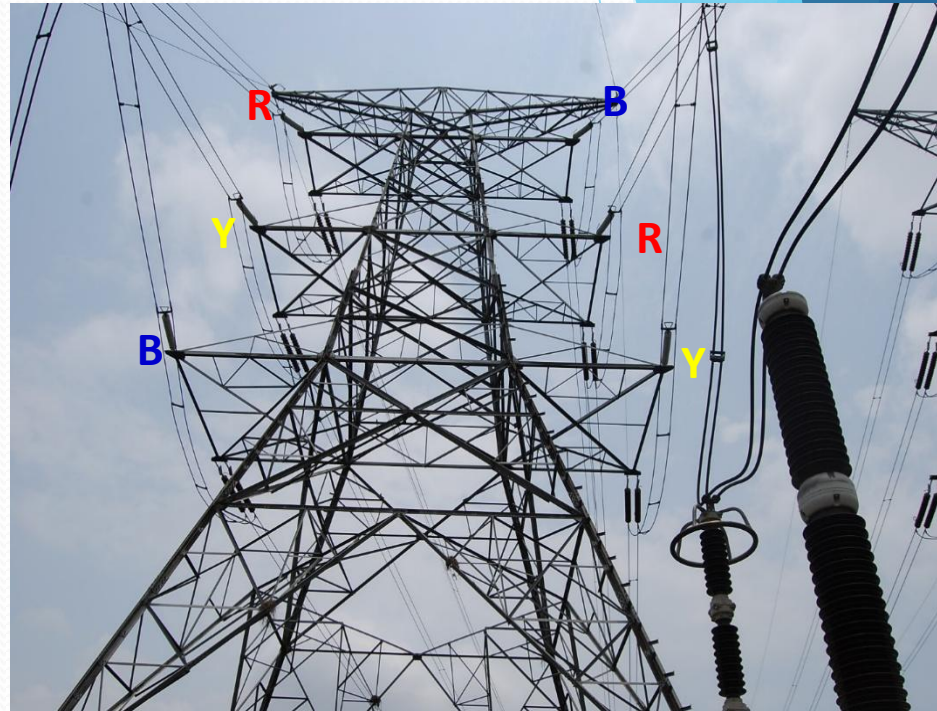
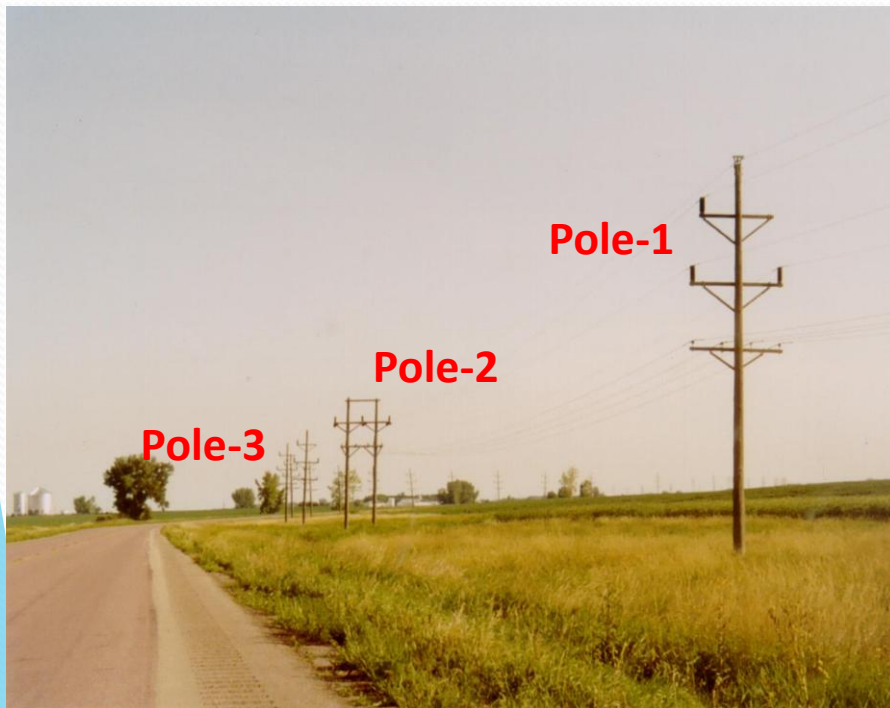
$$L_a = 2 \times 10^{-7} \ln \frac{D_{eq}}{D_s} \quad \text{H/m} \quad (4.6.18)$$

D_{eq} , the cube root of the product of the three-phase spacings, is the geometric mean distance between phases. Also, D_s is the conductor GMR for stranded conductors, or r' for solid cylindrical conductors.

Circuit Parameters of Overhead Transmission Lines

Inductance of a 3-phase line with equilateral spacing

INDUCTANCE OF 3-PHASE LINE WITH UNSYMMETRICAL SPACING



EXAMPLE OF LINE TRANSPOSITION

Circuit Parameters of Overhead Transmission Lines

Example#1

Single phase line operating at **50 Hz**. Conductor **spacing is 1m** and **diameter** of the conductor is **1.25cm**. Find inductive reactance of one conductor.

Solution:

1. Step 1

Calculate the GMR

- The GMR is given by $r' = 0.7788r$.
- Substitute the value of r :
- $r' = 0.7788 \times 0.00625 \text{ m} = 0.0048675 \text{ m}$.

2. Step 2

Calculate the inductance per meter

- The inductance per meter is given by $L = 2 \times 10^{-7} \ln\left(\frac{d}{r'}\right) \text{ H/m}$
- Substitute the values of d and r' :
- $L = 2 \times 10^{-7} \ln\left(\frac{1}{0.0048675}\right) \text{ H/m}$.
- $L = 2 \times 10^{-7} \ln(205.44) \text{ H/m}$.
- $L \approx 2 \times 10^{-7} \times 5.325 \text{ H/m}$.
- $L \approx 1.065 \times 10^{-6} \text{ H/m}$.

3. Step 3

Calculate the inductive reactance per meter

- The inductive reactance per meter is given by $X_L = 2\pi fL$.
- Substitute the values of f and L :
- $X_L = 2\pi \times 50 \text{ Hz} \times 1.065 \times 10^{-6} \frac{\text{H}}{\text{m}}$.

Solution

The inductive reactance of the single-phase line is approximately $3.346 \times 10^{-4} \frac{\Omega}{\text{m}}$.

Circuit Parameters of Overhead Transmission Lines

Practice#1

Single phase line **operating at 50 Hz**. Conductor is **Partridge** and **spacing is 6m** between centers.
Find inductive reactance of the line.

A typical GMR for a Partridge conductor is 0.0217 feet, according to reference. This is approximately 0.006614 meters.

The inductance of a conductor

$$L = (2 \times 10^{-7} \text{ H/m}) * \ln(6 / 0.006614)$$

$$L = (2 \times 10^{-7} \text{ H/m}) * \ln(907.17)$$

$$L = (2 \times 10^{-7} \text{ H/m}) * 6.8106$$

$$L \approx 1.268 \times 10^{-6} \text{ H/m}$$

The inductive reactance of a conductor= $X_L = 2\pi fL$.

Since the conductors composing the two sides of the line are identical, the inductive reactance of the line is: $2 \times X_L$

Circuit Parameters of Overhead Transmission Lines

Example#2

For $N=3$ and $M=2$, evaluate L_x , L_y , and L in H/m for the single-phase two-conductor line shown in the figure below.

SOLUTION For $N = 3$ and $M = 2'$, (4.6.6) becomes

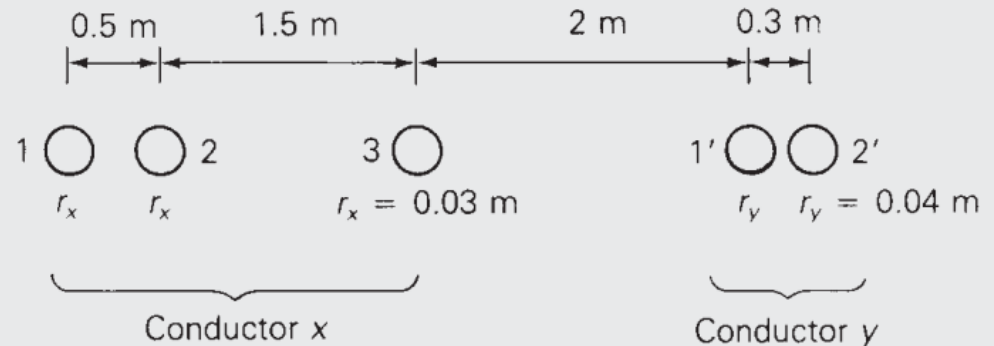
$$\begin{aligned} D_{xy} &= \sqrt[6]{\prod_{k=1}^3 \prod_{m=1'}^{2'} D_{km}} \\ &= \sqrt[6]{\prod_{k=1}^3 D_{k1'} D_{k2'}} \\ &= \sqrt[6]{(D_{11'} D_{12'})(D_{21'} D_{22'})(D_{31'} D_{32'})} \end{aligned}$$

Similarly, (4.6.7) becomes

$$\begin{aligned} D_{xx} &= \sqrt[9]{\prod_{k=1}^3 \prod_{m=1}^3 D_{km}} \\ &= \sqrt[9]{\prod_{k=1}^3 D_{k1} D_{k2} D_{k3}} \\ &= \sqrt[9]{(D_{11} D_{12} D_{13})(D_{21} D_{22} D_{23})(D_{31} D_{32} D_{33})} \end{aligned}$$

and (4.6.9) becomes

$$\begin{aligned} D_{yy} &= \sqrt[4]{\prod_{k=1'}^{2'} \prod_{m=1'}^{2'} D_{km}} \\ &= \sqrt[4]{\prod_{k=1'}^{2'} D_{k1'} D_{k2'}} \\ &= \sqrt[4]{(D_{1'1'} D_{1'2'})(D_{2'1'} D_{2'2'})} \end{aligned}$$



$$D_{11'} = 4 \text{ m} \quad D_{12'} = 4.3 \text{ m} \quad D_{21'} = 3.5 \text{ m}$$

$$D_{22'} = 3.8 \text{ m} \quad D_{31'} = 2 \text{ m} \quad D_{32'} = 2.3 \text{ m}$$

$$D_{xy} = \sqrt[6]{(4)(4.3)(3.5)(3.8)(2)(2.3)} = 3.189 \text{ m}$$

$$D_{11} = D_{22} = D_{33} = r'_x = e^{-1/4} r_x = (0.7788)(0.03) = 0.02336 \text{ m}$$

$$D_{21} = D_{12} = 0.5 \text{ m}$$

$$D_{23} = D_{32} = 1.5 \text{ m}$$

$$D_{31} = D_{13} = 2.0 \text{ m}$$

$$D_{xx} = \sqrt[9]{(0.02336)^3 (0.5)^2 (1.5)^2 (2.0)^2} = 0.3128 \text{ m}$$

$$D_{1'1'} = D_{2'2'} = r'_y = e^{-1/4} r_y = (0.7788)(0.04) = 0.03115 \text{ m}$$

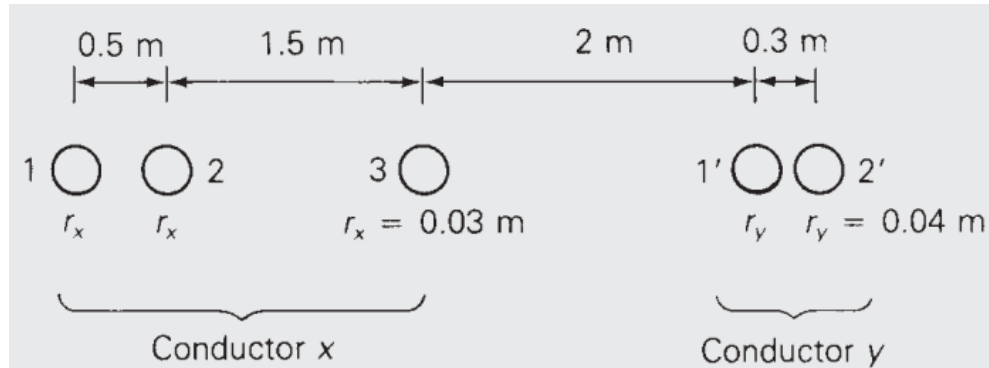
$$D_{1'2'} = D_{2'1'} = 0.3 \text{ m}$$

$$D_{yy} = \sqrt[4]{(0.03115)^2 (0.3)^2} = 0.09667 \text{ m}$$

Circuit Parameters of Overhead Transmission Lines

Example#2

For $N=3$ and $M=2$, evaluate L_x , L_y , and L in H/m for the single-phase two-conductor line shown in the figure below.



SOLUTION

Then, from (4.6.5), (4.6.8), and (4.6.10):

$$L_x = 2 \times 10^{-7} \ln \left(\frac{3.189}{0.3128} \right) = 4.644 \times 10^{-7} \text{ H/m per conductor}$$

$$L_y = 2 \times 10^{-7} \ln \left(\frac{3.189}{0.09667} \right) = 6.992 \times 10^{-7} \text{ H/m per conductor}$$

$$L = L_x + L_y = 1.164 \times 10^{-6} \text{ H/m per circuit}$$

It is seldom necessary to calculate GMR or GMD for standard lines. The GMR of standard conductors is provided by conductor manufacturers and can be found in various handbooks.

Also, if the distances between conductors are large compared to the distances between subconductors of each conductor, then the GMD between conductors is approximately equal to the distance between conductor centers.

Circuit Parameters of Overhead Transmission Lines

Example#3

A single-phase line operating at 60 Hz consists of two 4/0 12-strand copper conductors with 1.5 m spacing between conductor centers. The line length is 32 km. Determine the total inductance in H and the total inductive reactance in Ω .

SOLUTION The GMD between conductor centers is $D_{xy} = 1.5$ m. Also, from Table A.3, the GMR of a 4/0 12-strand copper conductor is $D_{xx} = D_{yy} = 0.01750$ ft or 0.5334 cm. From (4.6.5) and (4.6.8),

$$\begin{aligned} L_x = L_y &= 2 \times 10^{-7} \ln \left(\frac{150}{0.5334} \right) \frac{\text{H}}{\text{m}} \times 32 \times 10^3 \\ &= 0.03609 \quad \text{H per conductor} \end{aligned}$$

The total inductance is

$$L = L_x + L_y = 2 \times 0.03609 = 0.07218 \quad \text{H per circuit}$$

and the total inductive reactance is

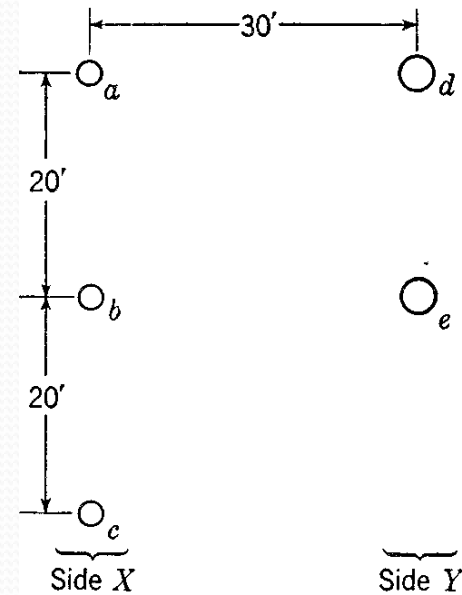
$$X_L = 2\pi fL = (2\pi)(60)(0.07218) = 27.21 \quad \Omega \text{ per circuit}$$



Circuit Parameters of Overhead Transmission Lines

Example#4 (Stevenson-4th edition)

One circuit of a **single-phase** transmission line is composed of three solid 0.1 in radius wires. The **return circuit** is composed of two 0.2 in radius wires. The arrangement of conductors is shown in the following figure. Find the inductance due to the current in each side of the line and the inductance of the complete line in millihenrys/mile.



Solution:

Find the GMD between sides X and Y.

$$\begin{aligned}
 D_m &= \sqrt[6]{D_{ad}D_{ae}D_{bd}D_{be}D_{cd}D_{ce}} \\
 D_{ad} &= D_{be} = 30 \text{ ft} \\
 D_{ae} &= D_{bd} = D_{ce} = \sqrt{20^2 + 30^2} = \sqrt{1,300} \\
 D_{cd} &= \sqrt{30^2 + 40^2} = 50 \text{ ft} \\
 D_m &= \sqrt[6]{30^2 \times 50 \times 1,300^{3/2}} \\
 &= 30^{1/3} \times 50^{1/6} \times 1,300^{1/4} = 35.8 \text{ ft}
 \end{aligned}$$

Then find the self GMD for side X.

$$\begin{aligned}
 D_s &= \sqrt[9]{D_{aa}D_{ab}D_{ac}D_{ba}D_{bb}D_{bc}D_{ca}D_{cb}D_{cc}} \\
 &= \sqrt[9]{\left(\frac{0.1 \times 0.7788}{12}\right)^3 \times 20^4 \times 40^2} \\
 &= \sqrt[3]{\frac{0.1 \times 0.7788}{12}} \times 20^{4/9} \times (2 \times 20)^{2/9} \\
 &= \sqrt[3]{\frac{0.1 \times 0.7788}{12}} \times 20^{2/3} \times 4^{1/6} = 1.605 \text{ ft}
 \end{aligned}$$

and for side Y

$$D_s = \sqrt[4]{\left(\frac{0.2 \times 0.7788}{12}\right)^2 \times 20^2} = 0.509 \text{ ft}$$

The inductance is, by Eq. (2.57),

$$\begin{aligned}
 L_X &= 0.7411 \log \frac{35.8}{1.605} = 1.00 \text{ millihenry/mile} \\
 L_Y &= 0.7411 \log \frac{35.8}{0.509} = 1.38 \text{ millihenrys/mile} \\
 L &= L_X + L_Y = 2.38 \text{ millihenrys/mile.}
 \end{aligned}$$

Circuit Parameters of Overhead Transmission Lines

Practice#2

A completely transposed 60-Hz three-phase line has flat horizontal phase spacing with 10 m between adjacent conductors. The conductors are 806 mm^2 (1,590,000 cmil) ACSR with 54/3 stranding. Line length is 200 km. Determine the inductance in H and the inductive reactance in Ω .

SOLUTION From Table A.4, the GMR of a 806 mm^2 (1,590,000 cmil) 54/3 ACSR conductor is

$$D_S = 0.0520 \text{ ft} \frac{1 \text{ m}}{3.28 \text{ ft}} = 0.0159 \text{ m}$$

Also, from (4.6.17) and (4.6.18),

$$D_{eq} = \sqrt[3]{(10)(10)(20)} = 12.6 \text{ m}$$

$$\begin{aligned} L_a &= 2 \times 10^{-7} \ln \left(\frac{12.6}{0.0159} \right) \frac{\text{H}}{\text{m}} \times \frac{1000 \text{ m}}{\text{km}} \times 200 \text{ km} \\ &= 0.267 \text{ H} \end{aligned}$$

The inductive reactance of phase a is

$$X_a = 2\pi f L_a = 2\pi(60)(0.267) = 101 \text{ } \Omega$$



Circuit Parameters of Overhead Transmission Lines

CAPACITANCE: SINGLE PHASE TWO WIRE LINE

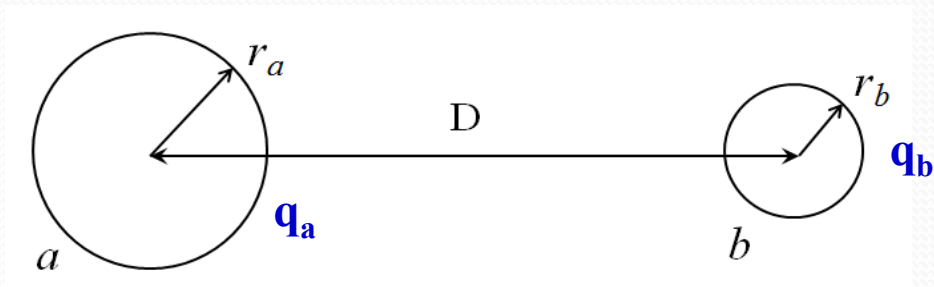
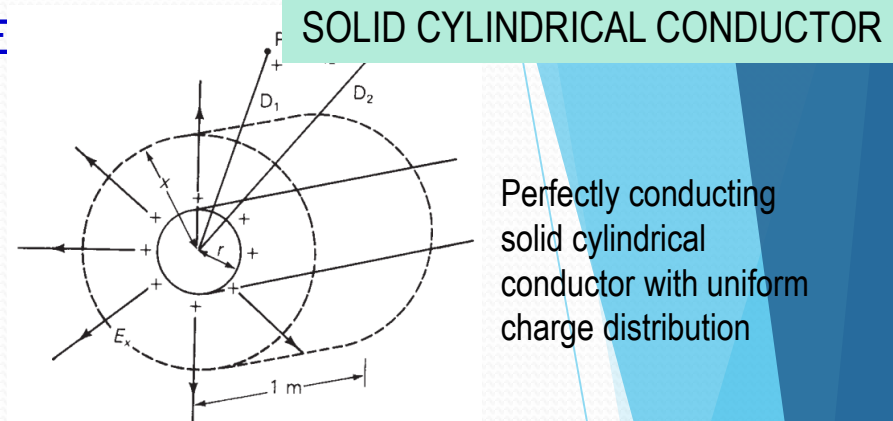


Figure: Cross-section of a parallel wire line.



Let a and b are the two conductors having radii r_a and r_b and charge q_a and q_b respectively. The capacitance between the conductors is,

$$C_{ab} = \frac{q_a}{V_{ab}} = \frac{2\pi k}{\ln \frac{D^2}{r_a \cdot r_b}} F/m \quad \text{If } r_a = r_b = r, \text{ then}$$

$$C_{ab} = \frac{\pi k}{\ln \frac{D}{r}} F/m$$

$$C_n = C_{an} = C_{bn} = \frac{q_a}{V_{ab/2}} = \frac{2\pi k}{\ln \frac{D}{r}} F/m \text{ to neutral}$$

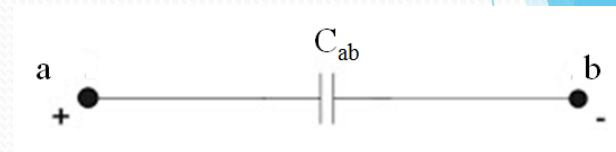


Figure: Representation of a line to line capacitance.

(Since $q_a = -q_b$)

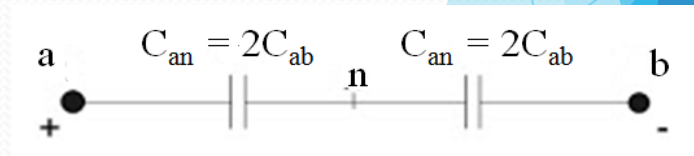


Figure: Representation of a line to neutral capacitance.

[assuming ϵ_r or $k_r = 1$, ϵ_0 or $k_0 = 8.85 \times 10^{-12} F/m$]

Circuit Parameters of Overhead Transmission Lines

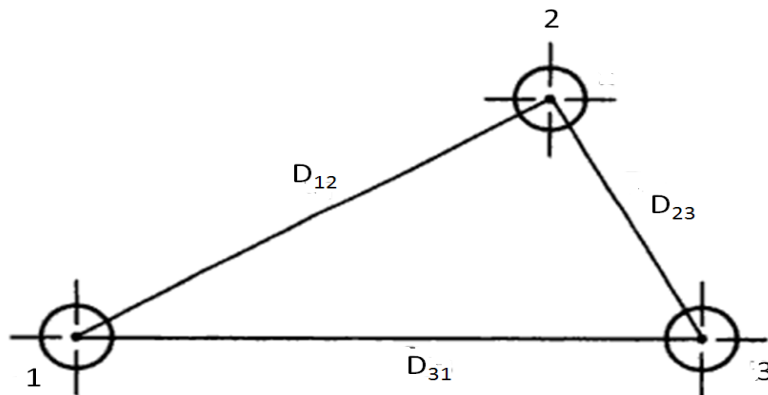
CAPACITANCE: THREE PHASE THREE WIRE LINE EQUAL/UNEQUAL SPACING

Capacitance to neutral,

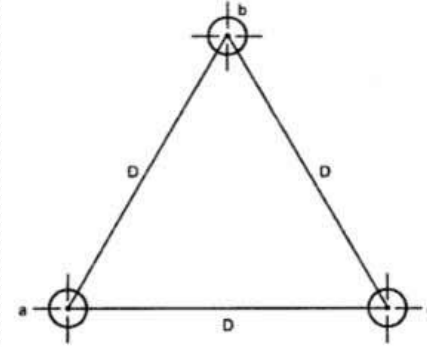
$$C_n = \frac{q_a}{V_{an}} = \frac{2\pi k}{\ln\left(\frac{D}{r}\right)} \frac{F}{m} \text{ to neutral}$$

$$C_{an} = \frac{q_a}{V_{an}} = \frac{2\pi k}{\ln \frac{D_{eq}}{r}} \text{ F/m to neutral}$$

Where, $D_{eq} = \sqrt[3]{D_{12}D_{23}D_{31}}$

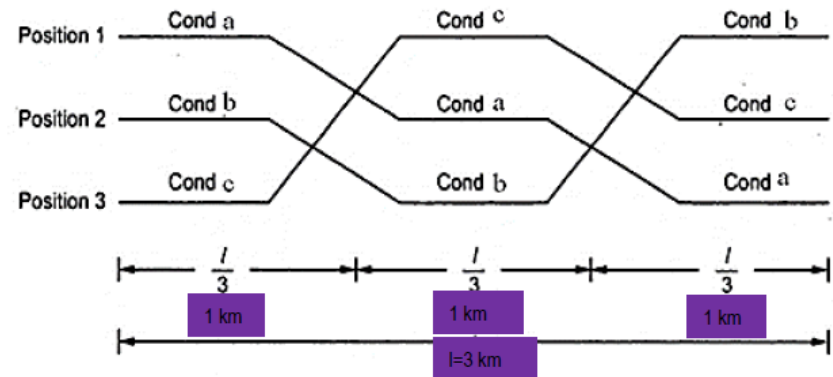


$q_a + q_b + q_c = 0$ in a balanced 3 phase circuit.



If there are no other charges in the vicinity, the sum of the charges on the three conductors is zero.

$$q_a + q_b + q_c = 0$$



Transposition
cycle

Circuit Parameters of Overhead Transmission Lines

Example#5

A single-phase line operating at 60 Hz consists of two 4/0 12-strand copper conductors with 1.5 m spacing between conductor centers. The line length is 32 km. determine the line-to-line capacitance in F and the line-to-line admittance in S. If the line voltage is 20 kV, determine the reactive power in kvar supplied by this capacitance.

SOLUTION From Table A.3, the outside radius of a 4/0 12-strand copper conductor is

$$r = \frac{0.552}{2} \text{ in.} = 0.7 \text{ cm}$$

and from (4.9.4),

$$C_{xy} = \frac{\pi(8.854 \times 10^{-12})}{\ln\left(\frac{150}{0.7}\right)} = 5.182 \times 10^{-12} \text{ F/m}$$

or

$$C_{xy} = 5.182 \times 10^{-12} \frac{\text{F}}{\text{m}} \times 32 \times 10^3 \text{ m} = 1.66 \times 10^{-7} \text{ F}$$

and the shunt admittance is

$$\begin{aligned} Y_{xy} &= j\omega C_{xy} = j(2\pi 60)(1.66 \times 10^{-7}) \\ &= j6.27 \times 10^{-5} \text{ S line-to-line} \end{aligned}$$

From (4.10.9),

$$Q_C = (6.27 \times 10^{-5})(20 \times 10^3)^2 = 25.1 \text{ kvar}$$



Circuit Parameters of Overhead Transmission Lines

Example#6

A 3-phase overhead transmission line has its conductors arranged at the corners of an equilateral triangle of 2 m side. Calculate the capacitance of each line conductor per km. Given that diameter of each conductor is 1.25 cm. Given that $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

Solution.

Conductor radius, $r = 0.625$ cm

Spacing of conductors, $d = 2$ m = 200 cm

$$\text{Capacitance of the line} = \frac{2\pi\epsilon_0}{\ln\frac{D}{r}} \text{ F/m}$$

$$= 0.0096 \times 10^{-9} \text{ F/m} = 0.0096 \times 10^{-6} \text{ F/km} = 0.0096 \text{ } \mu\text{F/km}$$

Practice#3

Calculate the capacitance of a 100 km long 3-phase, 50 Hz overhead transmission line consisting of 3 conductors, each of diameter 2 cm and spaced 2.5 m at the corners of an equilateral triangle. Given that $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

Circuit Parameters of Overhead Transmission Lines

Example#7

A 3-phase, 50 Hz, 132 kV overhead line has conductors placed in a horizontal plane 4 m apart. Conductor diameter is 2 cm. If the line length is 100 km, calculate the charging current per phase assuming complete transposition.

Solution.

Conductor radius, $r = 1$ cm

Now $d_1 = AB = 4$ m; $d_2 = BC = 4$ m; $d_3 = AC = 8$ m

$$\therefore D_{eq} = \sqrt[3]{d_1 \times d_2 \times d_3} = \sqrt[3]{4 \times 4 \times 8} = 5.04 \text{ m}$$

$$\text{Capacitance of the each phase to neutral} = \frac{2\pi\epsilon_0}{\ln \frac{D_{eq}}{r}} \text{ F/m}$$

$$= 0.00885 \times 10^{-6} \text{ F/km}$$

$$\text{Capacitance/phase for 100 km line is } C_n = 0.00885 \times 10^{-6} \times 100 = 0.885 \times 10^{-6} \text{ F}$$

$$\text{Charging current/phase, } I_C = 2\pi f C_n V_{ph} = (2\pi \times 50) \times (0.885 \times 10^{-6}) \times 76210 = 21.18 \text{ A}$$

$$V_{ph} = I_C X_C = I_C (1/2\pi f C_n)$$

$$V_{ph} = \text{Phase voltage} = \frac{132000}{\sqrt{3}} = 76210 \text{ V}$$

Circuit Parameters of Overhead Transmission Lines

Practice#4

A three-phase 60-Hz line has flat horizontal spacing. The conductors have an outside diameter of 3.28 cm with 12 m between conductors. Determine the capacitive reactance to neutral in ohm-meters and the capacitive reactance of the line in ohms if its length is 125 mi.

Solution:

$$D_{eq} = \sqrt[3]{12 \times 12 \times 24} = 15.12 \text{ m}$$

$$r = 0.0328/2 = 0.0164$$

$$X_C = \frac{2.862}{60} \times 10^9 \ln \frac{15.12}{0.0164} = 3.256 \times 10^8 \Omega \cdot \text{m}$$

For 125 miles,

$$X_C = \frac{3.256 \times 10^8}{125 \times 1609} = 1619 \Omega$$

Circuit Parameters of Overhead Transmission Lines

TABLE A.3 Characteristics of copper conductors, hard drawn, 97.3% conductivity

Size of Conductor		Num- ber of Strands	Diameter of Individual Strands (inches)	Outside Diam- eter (inches)	Breaking Strength (pounds)	Weight (pounds per mile)	Approx. Current Carrying Capacity* (amps)	Geometric Mean Radius at 60 Hz (feet)	r _s Resistance (Ohms per Conductor per Mile)							
Circular Mils	A.W.G. or B. & S.								25°C (77°F)				50°C (122°F)			
									dc	25 Hz	50 Hz	60 Hz	dc	25 Hz	50 Hz	60 Hz
1 000 000	...	37	0.1644	1.151	43 830	16 300	1 300	0.0368	0.0585	0.0594	0.0620	0.0634	0.0640	0.0648	0.0672	0.0685
900 000	...	37	0.1560	1.092	39 510	14 670	1 220	0.0349	0.0650	0.0658	0.0682	0.0695	0.0711	0.0718	0.0740	0.0752
800 000	...	37	0.1470	1.029	35 120	13 040	1 130	0.0329	0.0731	0.0739	0.0760	0.0772	0.0800	0.0806	0.0826	0.0837
750 000	...	37	0.1424	0.997	33 400	12 230	1 090	0.0319	0.0780	0.0787	0.0807	0.0818	0.0853	0.0859	0.0878	0.0888
700 000	...	37	0.1375	0.963	31 170	11 410	1 040	0.0308	0.0836	0.0842	0.0861	0.0871	0.0914	0.0920	0.0937	0.0947
600 000	...	37	0.1273	0.891	27 020	9 781	940	0.0285	0.0975	0.0981	0.0997	0.1006	0.1066	0.1071	0.1086	0.1095
500 000	...	37	0.1162	0.814	22 510	8 151	840	0.0260	0.1170	0.1175	0.1188	0.1196	0.1280	0.1283	0.1296	0.1303
500 000	...	19	0.1622	0.811	21 590	8 151	840	0.0256	0.1170	0.1175	0.1188	0.1196	0.1280	0.1283	0.1296	0.1303
450 000	...	19	0.1539	0.770	19 750	7 336	780	0.0243	0.1300	0.1304	0.1316	0.1323	0.1422	0.1426	0.1437	0.1443
400 000	...	19	0.1451	0.726	17 560	6 521	730	0.0229	0.1462	0.1466	0.1477	0.1484	0.1600	0.1603	0.1613	0.1619
350 000	...	19	0.1357	0.679	15 590	5 706	670	0.0214	0.1671	0.1675	0.1684	0.1690	0.1828	0.1831	0.1840	0.1845
350 000	...	12	0.1708	0.710	15 140	5 706	670	0.0225	0.1671	0.1675	0.1684	0.1690	0.1828	0.1831	0.1840	0.1845
300 000	...	19	0.1257	0.629	13 510	4 891	610	0.01987	0.1950	0.1953	0.1961	0.1966	0.213	0.214	0.214	0.215
300 000	...	12	0.1581	0.657	13 170	4 891	610	0.0208	0.1950	0.1953	0.1961	0.1966	0.213	0.214	0.214	0.215
250 000	...	19	0.1147	0.574	11 360	4 076	540	0.01813	0.234	0.234	0.235	0.235	0.256	0.256	0.257	0.257
250 000	...	12	0.1443	0.600	11 130	4 076	540	0.01902	0.234	0.234	0.235	0.235	0.256	0.256	0.257	0.257
211 600	4/0	19	0.1055	0.528	9 617	3 450	480	0.01668	0.276	0.277	0.277	0.278	0.302	0.303	0.303	0.303
211 600	4/0	12	0.1328	0.552	9 483	3 450	490	0.01750	0.276	0.277	0.277	0.278	0.302	0.303	0.303	0.303
211 600	4/0	7	0.1739	0.522	9 154	3 450	480	0.01579	0.276	0.277	0.277	0.278	0.302	0.303	0.303	0.303
167 800	3/0	12	0.1183	0.492	7 556	2 736	420	0.01559	0.349	0.349	0.349	0.350	0.381	0.381	0.382	0.382
167 800	3/0	7	0.1548	0.464	7 366	2 736	420	0.01404	0.349	0.349	0.349	0.350	0.381	0.381	0.382	0.382
133 100	2/0	7	0.1379	0.414	5 926	2 170	360	0.01252	0.440	0.440	0.440	0.440	0.481	0.481	0.481	0.481
105 500	1/0	7	0.1228	0.368	4 752	1 720	310	0.01113	0.555	0.555	0.555	0.555	0.606	0.607	0.607	0.607
83 690	1	7	0.1093	0.328	3 804	1 364	270	0.00992	0.699	0.699	0.699	0.699	0.765			

Reference:

Elements of Power System Analysis, W. G. Stevenson, 4th ed., McGraw Hill (Chapter 2)

Power System Analysis and Design, J. Duncan Glover, Mulukutla S. Sarma, and Thomas Overbye, 5th Edition. (Chapter 4)



QUESTIONS AND DISCUSSIONS

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